

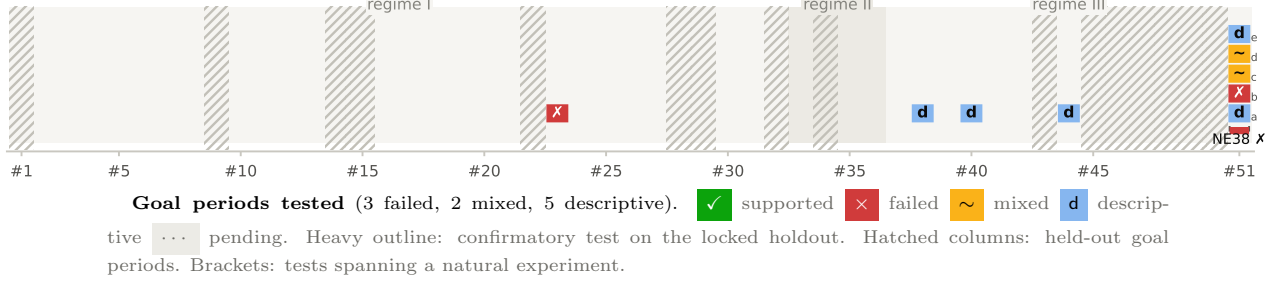
H22: Private, conflicting goals make #51 a spin glass

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 50% Faithfulness 1/5 Usefulness 1.5/5

Bottom line. #51 is not a spin glass: roles act as random fields with weak positive coupling, rivals align, and the swarm never switches collectively.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

Do the private, partly conflicting roles of #51 make the swarm a spin glass? Units 51a–e (45 non-holdout days) are tested against the shared-objective weeks #38, #40, #44. Content couplings are real but mostly positive, positions persist without collective switching ($W \approx 1$), and same-role rivals co-move *more* than unrelated pairs ($T_{SR} +0.04$ to $+0.06$ in every round-1b variant, $p \leq 0.05$). DQ2 stance is ferromagnetic. **Round 1c** uses the validated v2.1 disagreement flag with the labeller’s confusion built into the null. Rivals are not more disagreeing ($+0.49$ pp $[-0.68, +1.67]$), but the flag is too rare to decide (power 0.47 at odds ratio 3). A read vs in-flight contrast shows that reading a rival pulls content *toward* it ($\Gamma = +0.19 [+0.07, +0.31]$). The holdout is **not** run.

PHYSICAL MODEL

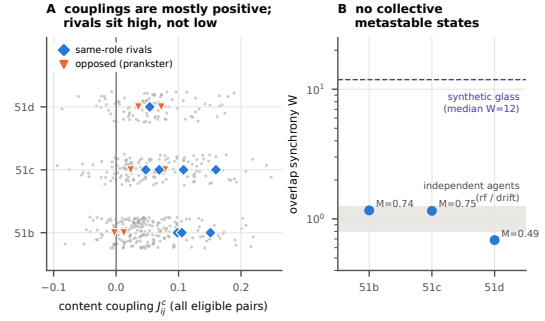
Spin glass (01, SK) of $O(n)$ spins (11). Spin $\mathbf{s}_i$: the whitened chat embedding of agent i per 30-min window ($n = 32$). Fields: day field $\mathbf{m}_d$; static field $\mathbf{H}_i$ (role, lab, style). J_{ij} : correlation of the within-day fluctuations $\mathbf{x}_i, \mathbf{x}_j$ minus a cross-day surrogate. A, B, C : day folds. κ : SK ratio J_0/J . τ_3 : gauge-invariant balance (SK ≈ 0 , factions ≈ 0.5). q : overlap of day-demeaned states $\hat{\delta}$. W : its synchrony beyond per-agent shifts.

$$P(\{\mathbf{s}\}) \propto \exp \left(\sum_{i < j} J_{ij} \mathbf{s}_i \cdot \mathbf{s}_j + \sum_i (\mathbf{m}_d + \mathbf{H}_i) \cdot \mathbf{s}_i \right)$$

$$J_{ij} = \text{corr}_w(\mathbf{x}_i, \mathbf{x}_j) - \text{corr}_w^{\text{surr}}(\mathbf{x}_i, \mathbf{x}_j)$$

$$\kappa = \sqrt{N} \bar{J} / \sigma_J, \quad \tau_3 = \text{tr}(J^A J^B J^C) / \text{tr}(J^A J^B)^{3/2}$$

$$q(d, d') = \frac{1}{N} \sum_i \hat{\delta}_{i,d} \cdot \hat{\delta}_{i,d'}$$



Observables. (a) The content coupling of every eligible pair in the counted #51 units (gray), with same-role rivals (blue) and Prankster vs Ethicist/Psychologist pairs (orange). Couplings are mostly positive, and rivals sit high. (b) Overlap synchrony W at the independent-agent level (gray band), far below a synthetic glass (dashed). Memory M is high without synchrony: agents hold their own positions, and the swarm never jumps between collective states.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Little changes in practice. Assigned roles act as fixed fields that pin each agent’s topic. The shared room adds a weak positive coupling, and same-objective agents drift together. To steer a mixed-motive swarm, edit the role prompts: roles, not inter-agent conflict, set what agents talk about. Expect no hidden collective states to escape ($W \approx 1$). Embeddings measure topic, not stance, so strategic opposition could stay invisible in content. The diagnostics port to other logs; the spin-glass vocabulary is not earned.

H22-private-goals-spin-glass · generated 2026-10-04 16:04 by infra/summaries/build_summaries.py · card hypotheses/H22-private-goals-spin-glass/README.md · ratings: RE-C2 round-1b agent (suggested; faithfulness and usefulness unchanged from the coordinator’s calibration)

RESULTS VS PREDICTIONS

Prediction	Round 1 → round 1b (51b / 51c / Verdict 51d)
P1 real couplings	content 2/3 → 2/3 (bge), 3/3 (gte); ✓ stance 3/3 beyond agent fields
P2 $\kappa < 1$	5.2 / 3.9 / 9.7; stance mean +0.6 ×
P3 $\tau_3^{\text{dc}} < 0.25$	content indeterminate (both models); stance 0.25 / 0.23 / 0.05 ✓
P4 rivals repel	RE $T_{\text{SR}} + 0.048 \rightarrow +0.042$ (gte), +0.052 × (ri- (style-res.); stance +0.09 n.s. val)
P4 opposed roles	content ≈ 0 ; stance +0.11 / -0.34 / × +0.22 (2 pairs)
P5 collective states	W 1.16 / 1.15 / 0.69; gte 51c 1.61 ($p \times 0.02$)
NE38 (native)	field leaves rivals ($p 0.004$); coupling DiD × -0.27 n.s.
#23 chess (na- tive)	opponents' stance -0.005; content × +0.03 n.s.
1c P4 stance (v2.1 flag)	DQ2 soft +0.09 n.s. → validated T_{SR}^D inconcl. +0.49 pp [-0.68, +1.67], NG $p 0.29$
1c P3 stance balance	3/3 → 1/3 ($\tau_3^{\text{dc}} 0.22 / 0.18 / 0.57$) ×
1c read vs in-flight	open → $\Gamma +0.19$ [+0.07, +0.31] (bge), rival +0.23 (gte)

GOAL PERIODS IN DETAIL

G23	× failed
G38	d 1b: descriptive (round 1: descriptive)
G40	d 1b: descriptive (round 1: descriptive)
G44	d 1b: descriptive (round 1: descriptive)
G51a	d 1b: descriptive (round 1: descriptive)
G51b	× 1b: failed (round 1: failed; rival homophily in every...
G51c	~ 1b: inconclusive (round 1: inconclusive)
G51d	~ 1b: inconclusive (round 1: inconclusive)
G51e	d 1b: descriptive (round 1: descriptive)
NE38	× failed

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	0
D unfitted predictions	1	I transfer	0
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

τ_3^{dc} cannot separate SK from a heterogeneous ferromagnet (Amendment 1), so the unbalanced stance residuals are not glass evidence. Per-pair negative-stance counts assume independent replies within a pair, so threads inflate them. They are not enriched in conflict classes. Each unit has few testable rival pairs (3–5; one in most shared splits). The NE38 coupling result uses a threshold chosen after seeing the coverage. DQ2 stance labels were reliable only in aggregate (pair-level “opposes” precision 0.08). The v2.1 flag is validated but rare (about 4 expected rival flags per unit), and its synthetic size is 0.11 in 51d. The Γ excess over random pair sets is suggestive only (percentile 0.73–0.95).

ROUND 2

Pool the validated flag over #51 and the holdout tail for power (re-freeze proposed). Fit lagged couplings per pair to see whether the read-gated rival pull is pair heterogeneity or role structure.

- **Essence**. Do conflicting incentives produce conflict behavior at all?
- **H22-R1** Cooperation default: agents help rivals at rates comparable to aligned pairs (Jev-labeled help acts).
- **H22-R2** A quantitative random-field fit (HH129): roles as fields plus weak positive coupling.

Second figure. (a) Rival contrast T_{SR} per unit in content (two models and style-residualized) and in stance; bars show the role-permutation sd. It is positive almost everywhere. The one negative stance value (pu51e, Opus 5 as game dev) is not significant. (b) NE38: the alignment and coupling of Opus 5 with its game-dev rivals while it held that role, and after.

